Provenance of 'dirac16complex in the primordial, modern and future gravitational fields: Euler-Lagrange equations, Kohn-Sham DFT, and pair creation'

1. Sources

- **Pre-gravity6plane.nb** (P. L. Nash, (C) 2022, GPL): the real Dirac matrices $T^A = (T16^A)[A]$ (cell 308), $\sigma_{16} = T^0T^1T^2T^3$ (cell 306), $\Gamma = (T16^A)[8]$ (cell 309); the primordial vielbein, which the notebook displays in text cells 901 and 913 as diag($\tan 6Hx_0$, $e^{-a_4(Hx_4)}\sin^{-1/6}6Hx_0$ (×3), 1, $e^{a_4(Hx_4)}\sin^{-1/6}6Hx_0$ (×3)); the assumptions $\cot 6Hx_0 > 0$, $\sin 6Hx_0 > 0$ (cell 48), i.e. $0 < 6Hx_0 < \pi/2$; the commented-out curved Lagrangian with $\sqrt{\det g}$ and spin connection (cells 1052, 1062).
- **User instructions**: Euler-Lagrange equations for interaction with the primordial, modern and future gravitational fields; a DFT-motivated approximation with a Kohn-Sham fermion-gas thermodynamic effective potential employing ground- and first-excited-state computations; proof that universes of masses $\{+m, -m\}$ are created in pairs for each case. The user did not specify the modern and future fields or the inflationary function a_4: the choices $a_4(Hx_4) = Hx_4$ (primordial), $f = 1 - r_s/x_0$ (modern) and $f = 1 - r_s/x_0 - H_\Lambda^2 x_0^2$ (future) are this work's definitions, chosen because the modern and future profiles possess static (Killing) horizons, the setting in which Einstein-Rosen bridges exist.
- **Standard physics and mathematics**: tetrad formalism and spin connection $\omega_\mu{}^A{}_B = e^A{}_\nu(\partial_\mu E_B{}^\nu + \Gamma^\nu{}_{\mu\lambda}E_B{}^\lambda)$; Kohn-Sham density-functional theory (W. Kohn and L. J. Sham, Phys. Rev. 140, A1133 (1965)) and its finite-temperature form (N. D. Mermin, Phys. Rev. 137, A1441 (1965)); the ΔSCF method for excited states; the Nambu-Jona-Lasinio mean-field gap equation (Y. Nambu and G. Jona-Lasinio, Phys. Rev. 122, 345 (1961)); MIT bag boundary conditions; the Lambert function W_0 ($x_0 = r_s(1 + W_0(e^{\rho/r_s - 1}))$ inverts $\rho = x_0 + r_s \ln(x_0/r_s - 1)$).

2. Computations (Wolfram Language 15.0.1, 2026-10-30)

result	where computed
covariant constancy $\partial_\mu\gamma^\nu + \Gamma^\nu{}_{\mu\lambda}\gamma^\lambda + [\Omega_\mu, \gamma^\nu] = 0$, sign of Ω_μ	6planes.nb Part 5, Out[23] (exact)
$\sqrt{\lvert g\rvert} = \sec 6Hx_0$, 1, 1	Part 5, Out[23]
reality $\mathcal{L}^* = \mathcal{L}$ and EL equations $= \sqrt{\lvert g\rvert}\sigma_{16}(\gamma^\mu D_\mu\psi - m\psi - \lambda(\bar\psi\psi)\psi)$ in all three fields	Part 5, Out[23] (exact Grassmann algebra, 16 components, 8 coordinates)
spin-connection terms $-3H\cot^2 6Hx_0$, $\frac{r_s}{4\sqrt{x_0^3(x_0 - r_s)}}$, $\frac{r_s - 2H_\Lambda^2 x_0^3}{4\sqrt{x_0^3(x_0 - r_s - H_\Lambda^2 x_0^3)}}$	Part 5, Out[23]
$\mathcal{L}_m[\Gamma\psi^*] = \mathcal{L}_{-m}[\psi]$, $\mathcal{L}_m[T^1\psi(R_1 x)] = \mathcal{L}_{-m}[\psi](R_1 x)$	Part 5.1, Out[24] (exact)
algebra of α, β, K, T^0	Part 6.1, Out[25] (exact)
bag test $1.13204, 1.47963, 1.95318$ vs exact $1.13211, 1.47986, 1.95361$	Part 6.2, Out[26]
table of Section 5 (V1, V2, ΔSCF)	Part 6.3, Out[27] (machine precision; self-consistency tolerance 10^{-11}; 51-55 iterations for V1)
$\Omega(T)$ table	Part 6.4, Out[29]
$\bar\psi\psi = S_+ - S_-$, $(\bar\psi\psi)^2 = S_+^2 + S_-^2 - 2S_+S_-$, 232 monomials in λS_+S_-, pair energy balance 0	Part 7, Out[30] (exact)

Independent prototypes of the curved-space and Kohn-Sham computations (scratch scripts grav_EL.wls and run_dft.wls) gave the same results before the notebook was assembled.

3. New in this work

The three gravitational fields as defined above; the complete Grassmann derivation of the curved Euler-Lagrange equations; the reduction of the Kohn-Sham problem to $h_{KS} = \alpha p + W\beta$ in tortoise coordinates with the exact ± symmetry and its staggered discretization; the two vacua V_1, V_2; Theorems 1-3 and their proofs.